The Impact of U.S. Monetary Policy on Oil Prices During the COVID-19 Pandemic: Evidence from 1990–2024

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December 9, 2024

Acknowledgements

I extend my deepest gratitude to Professor Zhining Hu and Professor Charles Weise for their invaluable guidance and feedback throughout this research. Professor Hu's assistance with econometric analysis contributed significantly to the robustness of the results, while Professor Weise's insights helped refine the theoretical framework and methodology.

Abstract

This study investigates the interaction between U.S. monetary policy and crude oil price dynamics during the COVID–19 pandemic. Using quarterly data from 1990 to 2024, the analysis incorporates lagged variables and interaction terms into an econometric framework designed to capture the changing transmission of monetary policy under supply-side stress. The results show that lagged GDP growth is the strongest predictor of crude oil price growth, while the direct effect of the federal funds rate is statistically weak. During the pandemic period, however, the interaction between lagged monetary policy and COVID-related disruptions becomes negative and economically large. In the refined Cochrane–Orcutt specification with semirobust standard errors, a 1 percentage point increase in the federal funds rate during the pandemic is associated with an estimated 9.995 percentage point decline in crude oil price growth. The findings suggest that monetary policy operates differently when inflation is shaped by supply constraints, logistics disruptions, and geopolitical shocks. The paper concludes that monetary tightening alone is not well suited to stabilize energy markets during supply-driven crises and should be complemented by fiscal, infrastructure, and energy-transition policies.

JEL Codes: E31, E52, Q41, C22

1 Introduction

The COVID–19 pandemic sent unprecedented shocks through international crude oil markets and exposed the vulnerability of energy systems to sudden global disruptions. On April 20, 2020, West Texas Intermediate (WTI) crude oil futures briefly fell to $-37 per barrel as demand collapsed during lockdowns and storage capacity tightened in hubs such as Cushing, Oklahoma (Energy Information Administration, 2020). At the same time, oil markets were shaped not only by weak demand, but also by supply-side pressures caused by transportation restrictions, labor shortages, and storage constraints. These dynamics created a setting in which the standard demand-channel view of monetary transmission may be incomplete.

To counter the broader economic downturn, the Federal Reserve cut the federal funds target range to 0.00%–0.25% and announced a $700 billion asset purchase program on March 15, 2020 (Federal Reserve, 2020). As the economy reopened, however, oil prices rebounded while supply frictions persisted. OPEC+ production cuts, bottlenecks at major U.S. ports, and later the Russia–Ukraine conflict all complicated the relationship between monetary policy and energy prices (OPEC, 2020; Reuters, 2021; IEA, 2022). By 2022, the Federal Reserve had shifted to aggressive tightening, raising rates to 4.25%–4.50% by year-end (Federal Reserve, 2022). Yet the effectiveness of monetary policy in the face of supply-driven inflation remained uncertain.

This paper studies whether U.S. monetary policy had a weaker effect on oil prices during the COVID–19 period than in more conventional macroeconomic environments. The empirical strategy uses quarterly data from 1990 to 2024 and estimates models that combine lagged macroeconomic variables with a COVID dummy and an interaction term between the pandemic period and the lagged federal funds rate. The paper therefore extends earlier work that emphasized short-run monetary transmission to energy markets (Rosa, 2013) and the decomposition of inflation into supply-driven and demand-driven components (Shapiro, 2022a,b, 2024).

The results show that GDP growth remains the most reliable predictor of crude oil price growth, while the direct coefficient on the lagged federal funds rate is not statistically significant in either the baseline or refined model. The interaction term, however, is negative and economically meaningful, suggesting that the pandemic period amplified the deflationary effect of tightening on oil price growth. In the refined Cochrane–Orcutt model with semirobust standard errors, the combined effect of the lagged federal funds rate and its interaction with the COVID period is -9.995, implying that a 1 percentage point increase in the federal funds rate during the pandemic is associated with an approximately 10 percentage point decline in oil price growth. This pattern suggests that monetary tightening interacted with disrupted supply conditions in a way that intensified downward pressure on oil prices rather than stabilizing inflation in the usual manner.

The rest of the paper proceeds as follows. Section 2 reviews the literature. Section 3 presents the theoretical framework and econometric model. Section 4 describes the data and transformations. Section 5 presents the regression results and diagnostic tests. Section 6 concludes with policy implications.

2 Literature Review

The relationship between monetary policy and oil prices has received substantial attention, but most of the literature focuses on how oil price shocks affect macroeconomic outcomes rather than how monetary policy feeds back into oil prices under different inflation regimes. Rosa (2013) provides a key reference point by showing that a 100-basis-point increase in the federal funds rate lowers oil prices by roughly 3%, largely through the exchange-rate channel. Tighter U.S. policy strengthens the dollar, which makes dollar-denominated oil more expensive for foreign buyers and dampens global demand. However, Rosa's framework focuses on short-run responses and does not distinguish between demand-driven and supply-driven inflationary environments.

More recent work by Shapiro (2022a,b, 2024) is especially relevant because it separates inflation into supply-driven and demand-driven components. That distinction matters here because pandemic-era inflation was shaped heavily by disrupted production, shipping, labor supply, and energy logistics. Shapiro's framework implies that monetary tightening is less effective when inflation is dominated by supply-side constraints. This paper extends that logic by examining crude oil prices directly and by modeling how the effect of the federal funds rate changes during the COVID period.

Other contributions help motivate the broader macroeconomic setting. Mahedy and Shapiro (2017) and Barnichon and Shapiro (2022) emphasize labor market conditions and inflation dynamics, while IMF (2020) argues that fiscal tools are essential during crises when monetary policy cannot resolve structural disruptions in energy markets. These studies support the idea that traditional interest-rate policy may be too blunt when the source of inflation lies outside aggregate demand.

Natural resource economics adds a second theoretical lens. In Hotelling's classic framework, the expected return on an exhaustible resource must move with the interest rate, so higher rates lower the present value of future extraction and can encourage producers to extract sooner (Hotelling, 1931). Recent work continues to support this intuition. Ekeland et al. (2022) show that higher interest rates can accelerate extraction in competitive resource markets, and the Federal Reserve Bank of Minneapolis explains the same basic mechanism in accessible terms (Federal Reserve Bank of Minneapolis, 2014). This paper contributes by bringing that resource-economics logic together with macroeconomic models of monetary transmission and inflation decomposition.

3 Theoretical Framework and Econometric Model

3.1 Theoretical Framework

Monetary policy can affect crude oil prices through at least three channels: aggregate demand, exchange rates, and market expectations. First, higher interest rates reduce borrowing, investment, and spending, which lowers aggregate demand for energy. Second, tighter monetary policy tends to appreciate the U.S. dollar, which makes oil more expensive for foreign buyers and can reduce global demand. Third, policy actions affect expectations in futures markets and may dampen speculative demand for oil. These mechanisms are central to the monetary-policy channel described by Rosa (2013).

During the COVID–19 period, however, these channels interacted with unusual supply-side disturbances. Port congestion, transportation bottlenecks, labor shortages, OPEC+ production decisions, and geopolitical shocks all constrained energy supply even as demand conditions evolved rapidly. Under these conditions, monetary policy may have had a weaker direct effect on inflation but still affected oil prices indirectly through financial conditions, dollar strength, and expectations. This logic is consistent with the supply-driven inflation framework developed by Shapiro (2024).

Hotelling's theory adds another mechanism. When interest rates rise, the present value of future oil revenue falls, which can encourage more immediate extraction and increase current supply (Hotelling, 1931; Ekeland et al., 2022; Federal Reserve Bank of Minneapolis, 2014). Thus, the total effect of monetary tightening on oil prices may reflect both weaker demand and changes in extraction incentives.

3.2 Econometric Model

The paper estimates the relationship between crude oil price growth and macroeconomic conditions using quarterly data from 1990 to 2024. The main empirical model is

$$\begin{aligned} dcoilwtico_growth_t = \beta_0 + \beta_1 gdpc1_growth_lag_t + \beta_2 fedfunds_lag_t \\ + \beta_3 cpaltt01usm657n_growth_lag_t + \beta_4 covid_t + \beta_5 inter_t + u_t \end{aligned} \tag{1}$$

Here, *dcoilwtico_growth* is the growth rate of WTI crude oil prices, *gdpc1_growth_lag* is lagged real GDP growth, *fedfunds_lag* is the lagged federal funds rate, and *cpaltt01usm657n_growth_lag* is lagged CPI growth. The variable *covid* is a dummy equal to one during the pandemic window, and *inter* is the interaction term defined by

$$inter_t = covid_t \times fedfunds_lag_t. \tag{2}$$

Because the marginal effect of the lagged federal funds rate depends on whether the economy is in the pandemic period, the total effect of monetary policy during COVID is

$$\frac{\partial dcoilwtico_growth_t}{\partial fedfunds_lag_t} = \beta_2 + \beta_5 \cdot covid_t. \tag{3}$$

When $covid_t = 1$, the relevant marginal effect becomes $\beta_2 + \beta_5$. This follows the standard interpretation of interaction terms in regression analysis (Wooldridge, 2020).

To address first-order serial correlation, the paper also estimates a Cochrane–Orcutt AR(1) specification using `prais, corc robust`. This refinement preserves the same right-hand-side variables but adjusts the error process and reports semirobust standard errors.

4 Data and Variable Construction

The analysis uses quarterly data from 1990 to 2024. The variables come from the FRED-based dataset used in the Stata workflow. The Stata log shows the exact transformations used to construct the final dataset, including log transformations, lag creation, growth-rate calculations, the COVID dummy, and the interaction term (Stata Log, 2024, see also the uploaded Stata log).

An index variable was first created and set as the time variable in Stata. The log then generated logged levels of crude oil prices, real GDP, and CPI, followed by lagged versions of each series. Growth variables were defined as 100 times the log difference between the current value and its lag. In Stata notation, for crude oil prices this took the form

$$dcoilwtico_growth_t = 100 \times [\ln(dcoilwtico_t) - \ln(dcoilwtico_{t-1})]\,. \tag{4}$$

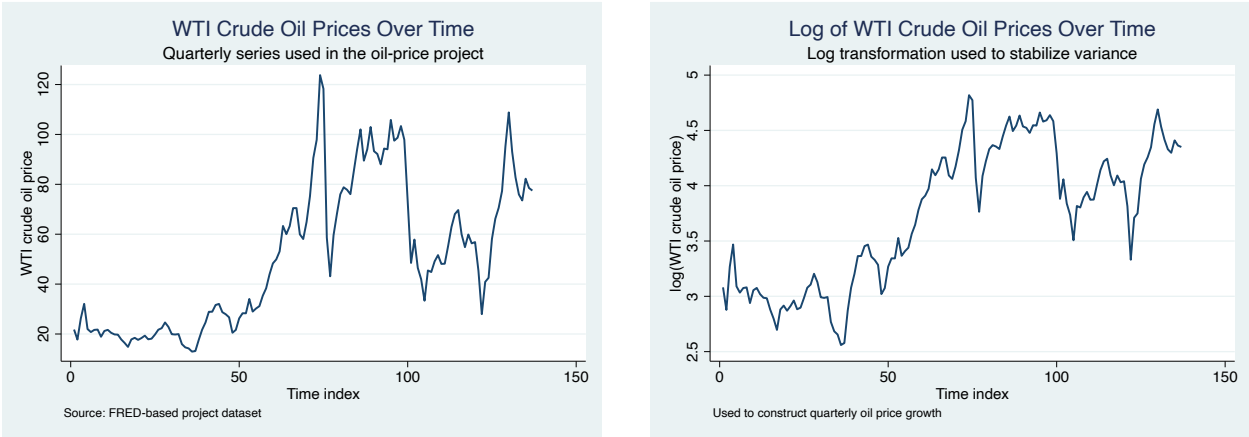
Analogous transformations were applied to GDP and CPI. The COVID dummy was coded as one for observations with index values above 117 and below 134, which corresponds to the pandemic-disruption window in the dataset. The interaction term was then generated as the product of the COVID dummy and the lagged federal funds rate. These steps are documented directly in the Stata log (Stata Log, 2024).

The data transformations serve three purposes. First, log transformations reduce heteroskedasticity and allow percentage-change interpretations. Second, lagged variables better reflect the delayed transmission of monetary policy. Third, the interaction term lets the effect of policy vary across regimes. These transformations follow standard econometric practice in macroeconomic time-series work (Wooldridge, 2020; Bernanke, Gertler, and Watson, 1997).

Table 1. Summary of Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
dcoilwtico_growth	0.93	15.71	-70.10	38.90
gdpc1_growth_lag	0.61	1.14	-8.24	7.54
fedfunds_lag	2.77	2.37	0.06	8.25
cpaltt01usm657n_growth_lag	-19.77	164.45	-673.50	626.56
covid (dummy)	0.12	0.32	0.00	1.00
inter	0.13	0.52	0.00	3.65

Table 1 highlights the extreme volatility of oil price growth and CPI growth relative to GDP growth and the federal funds rate. The large negative minimum for oil price growth reflects the 2020 collapse, while the positive maximum reflects the rebound that followed. These patterns are consistent with the paper's emphasis on crisis-period instability.



(a) Level of WTI crude oil prices.

(b) Log of WTI crude oil prices.

Figure 1. Crude oil prices before and after log transformation.

Figure 1 illustrates why the log transformation is useful. The level series displays large swings and changing variance, while the log series is smoother and easier to interpret in percentage-change terms.

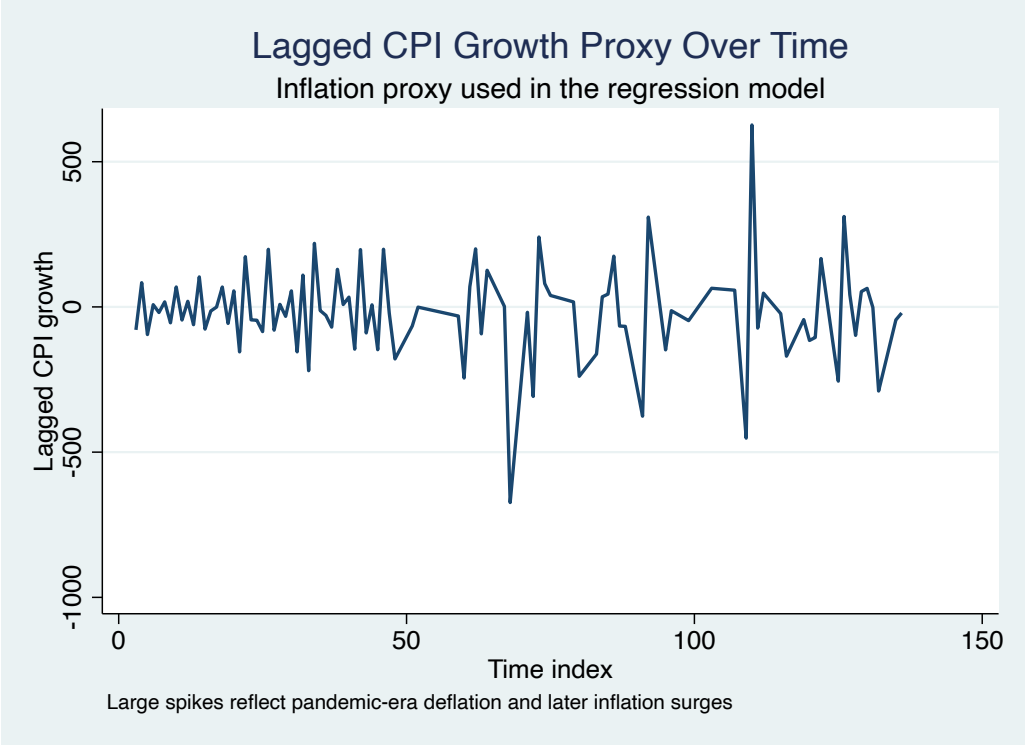


Figure 2. Lagged CPI growth proxy over time.

Figure 2 tracks the inflation proxy used in the regressions. The large negative and positive spikes underscore the instability of price dynamics during the pandemic and the later inflation surge.

5 Results

5.1 Baseline Results

The initial OLS regression uses 97 observations and explains about 18.0% of the variation in crude oil price growth. The lagged GDP growth coefficient is positive and statistically significant at the 1% level, while the lagged federal funds rate is small and statistically insignificant. The COVID dummy is also insignificant, but the interaction term between the pandemic period and the lagged federal funds rate is negative and significant. These results are consistent with the values reported in the Stata log (Stata Log, 2024).

Table 2. Regression Results Before Model Refinement

Variable	Coefficient	Std. Error	t-Statistic	P-value	[95% Conf. Interval]
`gdpc1_growth_lag`	7.014	2.328	3.01	0.003	[2.390, 11.638]
`fedfunds_lag`	-0.074	0.599	-0.12	0.902	[-1.263, 1.116]
`cpaltt01usm657n_growth_lag`	0.003	0.008	0.34	0.734	[-0.014, 0.019]
`covid`	7.503	5.790	1.30	0.198	[-3.998, 19.003]
`inter`	-11.364	4.069	-2.79	0.006	[-19.447, -3.282]
`_cons`	-2.678	2.893	-0.93	0.357	[-8.426, 3.069]
Model Statistics					
Number of obs			97		
F(5, 91)			4.00		
Prob > F			0.0025		
R-squared			0.1801		
Adj R-squared			0.1351		
Root MSE			12.653		

The baseline estimates imply that GDP growth is the strongest short-run correlate of oil price growth. A 1 percentage point increase in lagged GDP growth is associated with about a 7 percentage point increase in crude oil price growth. By contrast, the direct effect of the lagged federal funds rate is close to zero. The negative interaction coefficient suggests that policy tightening had a more negative association with oil prices during the COVID period than outside it.

5.2 Diagnostic Tests

The Gauss–Markov diagnostics provide mixed support for the baseline OLS specification. The Ramsey RESET test does not reject the null of correct functional form, which suggests no strong evidence of omitted nonlinear structure. Multicollinearity is also limited, as the mean VIF is only 1.48. The more serious issues are heteroskedasticity and likely serial correlation. The Breusch–Pagan/Cook–Weisberg test rejects homoskedasticity, and the Durbin–Watson statistic of 1.2771 points to positive serial correlation. These diagnostics are all recorded in the Stata log (Stata Log, 2024).

These results motivate a refined specification that corrects for both heteroskedasticity and serial dependence.

Table 3. Summary of Statistical Tests

Test	Statistic	p-value
Ramsey RESET test	$F(3, 88) = 1.89$	0.1370
Breusch–Pagan/Cook–Weisberg test	$\chi^2(1) = 9.60$	0.0019
Durbin–Watson test	$d = 1.2771$	—
Variance Inflation Factor	Mean VIF = 1.48	—
VIF Details		
`covid`	VIF = 2.20, 1/VIF = 0.454	
`inter`	VIF = 1.97, 1/VIF = 0.508	
`fedfunds_lag`	VIF = 1.19, 1/VIF = 0.839	
`cpaltt01usm657n_growth_lag`	VIF = 1.03, 1/VIF = 0.973	
`gdpc1_growth_lag`	VIF = 1.02, 1/VIF = 0.977	

5.3 Refined Model

The paper next applies robust standard errors and then a Cochrane–Orcutt AR(1) correction. The final reported model uses `prais, corc robust`, which yields 81 observations after transformation and produces a Durbin–Watson statistic of 1.8006 in the transformed regression. That value is much closer to 2, which suggests the autocorrelation problem is substantially reduced (Stata Log, 2024).

Table 4. Regression Results After Model Refinement (Cochrane–Orcutt, Semirobust)

Variable	Coefficient	Std. Error	t-Statistic	P-value	[95% Conf. Interval]
`gdpc1_growth_lag`	7.951	3.209	2.48	0.015	[1.558, 14.343]
`fedfunds_lag`	-0.800	0.838	-0.95	0.343	[-2.470, 0.871]
`cpaltt01usm657n_growth_lag`	0.007	0.007	1.00	0.318	[-0.007, 0.021]
`covid`	0.656	6.461	0.10	0.919	[-12.215, 13.526]
`inter`	-9.195	5.053	-1.82	0.073	[-19.261, 0.872]
`_cons`	-1.940	3.833	-0.51	0.614	[-9.574, 5.696]
Model Statistics					
Number of obs			81		
F(5, 75)			2.86		
Prob > F			0.0202		
R-squared			0.2167		
Root MSE			11.593		
Additional Statistic					
ρ			0.16819		

The refined model preserves the main substantive result: GDP growth remains significant, while the direct coefficient on the lagged federal funds rate remains insignificant. The interaction term becomes somewhat less precisely estimated but remains economically large and negative. Using equation (3), the total effect of monetary policy during the pandemic is

$$\hat{\beta}_2 + \hat{\beta}_5 = -0.800 + (-9.195) = -9.995. \tag{5}$$

This implies that during the COVID period, a 1 percentage point increase in the federal funds rate is associated with an estimated 9.995 percentage point decline in crude oil price growth. The result does not mean that policy alone determined oil prices; rather, it suggests that monetary tightening operated in a crisis environment where supply disruptions amplified the observed downturn in oil price growth.

5.4 Comparing the Initial and Refined Models

Two findings are especially stable across specifications. First, lagged GDP growth is consistently positive and statistically significant. This supports the standard view that stronger real activity raises energy demand and therefore supports higher oil prices (Hamilton, 2009). Second, the lagged federal funds rate by itself is not statistically different from zero in either model. That pattern is consistent with the argument that monetary policy affects oil markets indirectly and that its direct predictive power is weaker during supply-driven episodes.

The main difference across models concerns the precision and interpretation of the interaction term. In the baseline OLS model, the interaction term is statistically significant at conventional levels. In the refined model, its p-value rises to 0.073, so it is only marginally significant. Still, the coefficient remains economically important and negative. The refined model also improves the overall fit modestly, raising the R-squared from 0.1801 to 0.2167 and reducing the severity of serial correlation.

6 Conclusion

This paper examined how U.S. monetary policy related to crude oil price dynamics during the COVID–19 pandemic. Using quarterly data from 1990 to 2024, the analysis combined lagged macroeconomic variables with a COVID dummy and an interaction term to test whether the transmission of monetary policy changed in a crisis dominated by supply-side disruptions.

The results point to three main conclusions. First, lagged GDP growth is the strongest and most consistent predictor of crude oil price growth, which highlights the importance of real activity and energy demand. Second, the direct effect of the lagged federal funds rate is weak and statistically insignificant. Third, the interaction between the pandemic period and the lagged federal funds rate is negative and economically meaningful, suggesting that monetary tightening had a stronger deflationary association with oil prices during COVID than in normal times.

These findings support the view that monetary policy is less effective as a stabilization tool when inflation is driven by supply bottlenecks, logistics disruptions, and geopolitical shocks rather than excess demand. In that setting, complementary fiscal and structural policies matter more. Policies that improve supply chain resilience, expand energy infrastructure, and accelerate diversification toward renewable energy are likely to be more effective complements to monetary tightening during future crises (IMF, 2020; European Commission, 2022; White House, 2024).

The study also has limitations. The sample is relatively small once transformed and adjusted for

autocorrelation, and the model explains only a modest share of the variation in crude oil price growth. Future work could use higher-frequency data, richer supply-side controls, or alternative identification strategies to separate demand effects from geopolitical and production shocks more cleanly.

Figure Placeholders and Suggested Captions

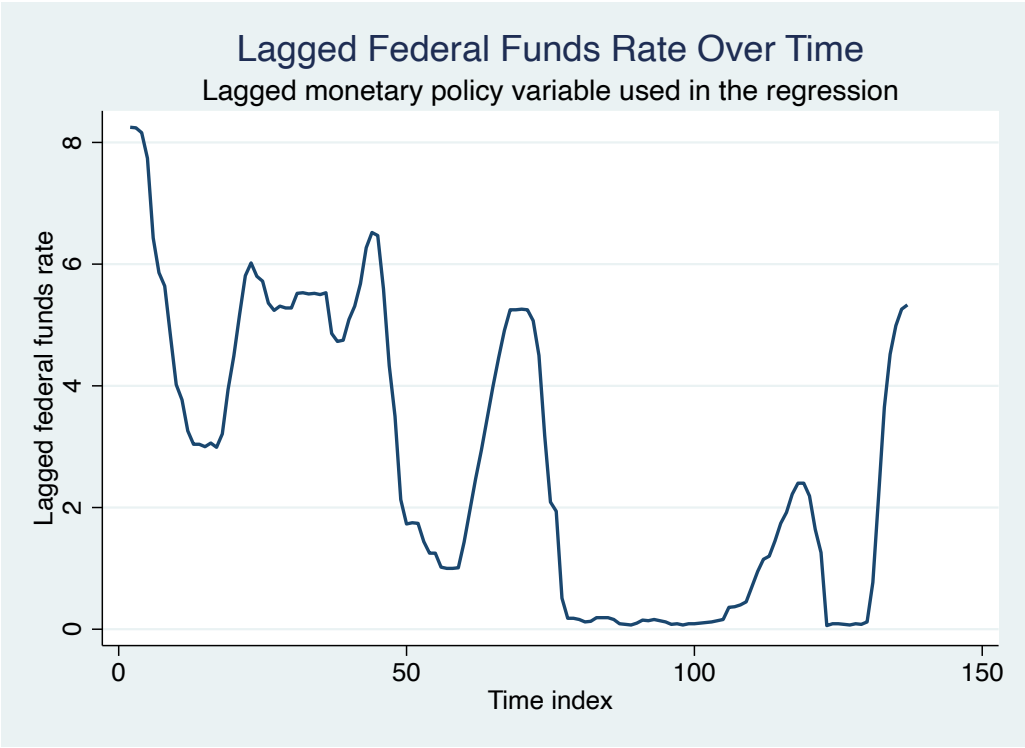


Figure 3. Lagged federal funds rate over time.

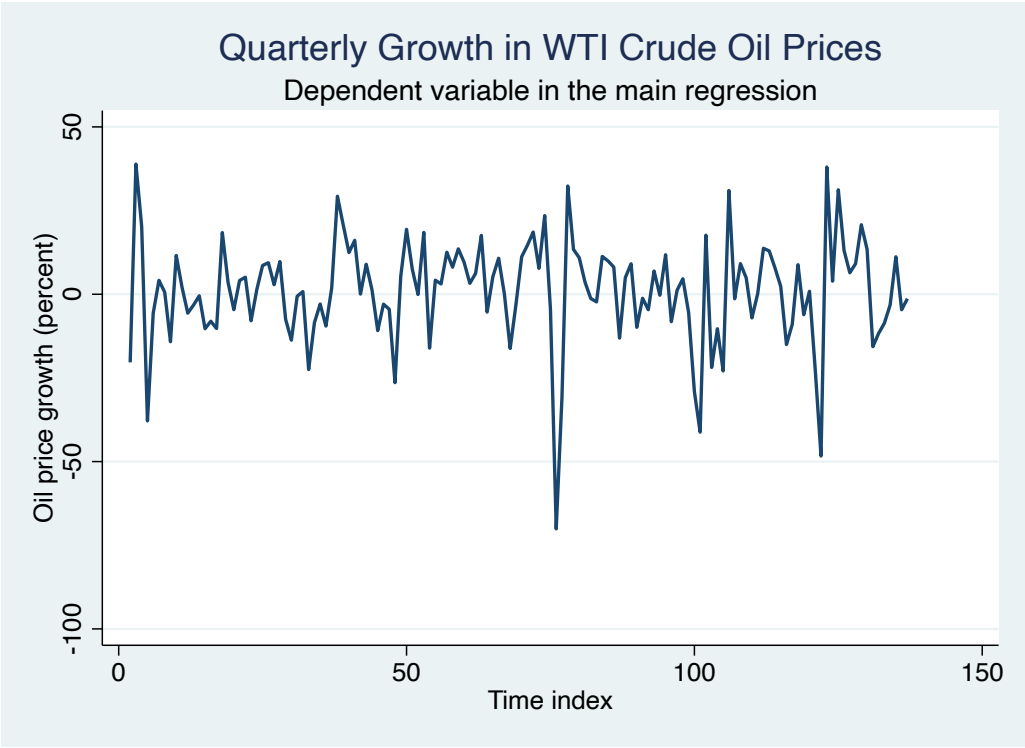


Figure 4. Quarterly crude oil price growth over time.

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Final Stata .log. User-uploaded Stata log file used to generate the empirical results and graph workflow for this project.